TryHackMe_What_is_Networking.md

This room provides a foundational overview of networking principles, explaining how computing devices interconnect, how the internet functions, how network devices are identified, and how basic diagnostics like ICMP Ping work.

Task 1: What is Networking?

Concept: A network is formed whenever two or more devices connect to share resources and communicate. In modern computing, networks span everything from personal devices to critical power grids and smart infrastructure.

Key Terminology: Individual computing devices within a network are categorized as network devices or nodes.

Task 2: What is the Internet?

Concept: The internet is a global system of interconnected computer networks consisting of private and public networks.

ARPANET: Funded by the US Department of Defense in the late 1960s, it served as the earliest precursor to the modern internet.

World Wide Web (WWW): Invented in 1989 by Tim Berners-Lee, enabling information storage, retrieval, and sharing across global networks.

Questions & Answers:

Question: Who invented the World Wide Web? \rightarrow Tim Berners-Lee

Task 3: Identifying Devices on a Network

Concept: Devices on a network identify themselves using logical and physical addressing schemes.

IP Address (Logical): Standard logical network identifier. IPv4 addresses consist of 4 octets/sections separated by dots (e.g., 86.157.52.21).

MAC Address (Physical): Media Access Control address—a unique 12-character hexadecimal physical address assigned to the Network Interface Card (NIC) at the factory. The first 6 characters represent the vendor/manufacturer, while the last 6 characters represent the unique device identifier.

MAC Spoofing: A technique where a device impersonates another MAC address to bypass network access controls (such as hotel Wi-Fi paywalls or MAC filtering).

Questions & Answers:

Question 1: What does the term "IP" stand for? \rightarrow Internet Protocol
Question 2: What is each section of an IP address called? \rightarrow Octet
Question 3: How many sections (in digits) does an IPv4 address have? \rightarrow 4
Question 4: What does the term "MAC" stand for? \rightarrow Media Access Control
Question 5: Deploy the interactive lab... and spoof your MAC address to access the site. What is the flag? \rightarrow THM{...} (Extracted upon changing Bob's MAC address to match Alice's in the interactive hotel Wi-Fi simulation)

Task 4: Ping (ICMP)

Concept: Ping is a fundamental network diagnostic tool that utilizes the Internet Control Message Protocol (ICMP) to verify reachability and measure round-trip latency between devices.

Mechanism: Sends an ICMP Echo Request packet and waits for an ICMP Echo Reply from the destination.
Syntax: ping <IP_or_Domain> (e.g., ping 192.168.1.254).

Questions & Answers:

Question 1: What protocol does ping use? \rightarrow ICMP
Question 2: What is the syntax to ping 10.10.10.10? \rightarrow ping 10.10.10.10
Question 3: What flag do you get when you ping 8.8.8.8? \rightarrow THM{...} (Generated inside the simulated lab shell upon running ping 8.8.8.8)